

医药翻译PRD.pdf - 纯译文

第 1 段

【中文 -> 英文】

医药翻译PRD

原型参考文件持续更新中<http://10.3.75.127:3000/>

一、项目背景

1.1 业务背景

医药行业在日常研发、注册申报、学术交流及国际化合作中，涉及大量跨语言文档处理需求。典型场景包括：

- 药品临床试验报告、研究者手册、知情同意书等中英文文档互译；
- 医疗器械技术文件、说明书、标签的多语言版本制作；
- 医药学术论文、FDA/EMA/ NMPA 申报资料的快速翻译与对照审核。

目前，用户主要依赖通用翻译工具或人工翻译。通用工具缺乏医药专业术语库支持，导致关键术语翻译不准确、前后不一致；人工翻译成本高、周期长，无法满足高频、小批量的即时翻译需求。此外，现有工具对 Word、PDF 文档的格式保留能力不足，翻译后需大量人工调整排版，严重影响效率。

1.2 用户痛点

痛点 描述

术语不准确 通用翻译引擎无法识别 “ randomized controlled trial ” “ primary endpoint ” “ adverse event ” 等医药专业术语，翻译结果不可用

格式易破坏 PDF 或 Word 翻译后，表格、页眉页脚、图表布局错乱，需要大量人工修复

缺乏对照版本 翻译后仅有译文，无法快速与原文对照审核，双语版本需人工拼接

历史无法追溯 每次翻译需重新上传、设置语言，无法查阅之前的翻译结果

当前为本地开发占位译文。接入通义千问配置后可替换为真实模型翻译结果。

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Multilingual configuration is cumbersome—source and target languages must be manually selected each time, resulting in low efficiency.

Data security concerns—pharmaceutical documents contain sensitive information, and using public translation tools poses a risk of data leakage.

Inconvenience in handling multiple files—after uploading several documents at once, there is no unified management or side-by-side preview capability.

1.3 Technical Feasibility

- A hybrid approach combining machine translation and a pharmaceutical terminology database has been maturely applied in the industry, ensuring high accuracy of specialized terms.
- Document format preservation technologies (such as Apache POI, Aspose, and PDF parsing engines) enable stable layout output for Word/PDF files.
- Single Sign-On (SSO) combined with user-space isolation solutions meet enterprise-level security requirements.

2. Product Value

2.1 User Value

Value Point | Description

Accurate and Consistent Terminology | Built-in pharmaceutical terminology database ensures consistent use of terms throughout the document and compliance with industry standards—ready for official documentation.

Zero Format Adjustment | Translated Word/PDF files retain original formatting (allowing minor length adjustments while preserving direction and structure)—no manual corrections required.

Significant Efficiency Improvement | Instant text translation response; batch processing for files; one-click reuse of historical records—average translation time reduced by over 80%.

Convenient Side-by-Side Review | Automatically generates left-right comparison previews for translated documents, offering both pure translation and bilingual versions for download to meet different scenarios.

Full Traceability | All translation tasks are automatically saved, supporting search, filtering, and quick re-execution—preventing redundant work.

Secure Data Isolation | Single Sign-On + user data isolation ensures confidentiality and compliance of pharmaceutical documents.

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Intuitive operation — Task-oriented interface, clear at a glance, no training required to get started.

2.2 Business Value

Value Point | Description

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Reduce translation costs | Machine translation replaces most manual translation, reducing outsourcing expenses; estimated annual cost reduction of 60% – 70%.

Accelerate business processes | Real-time translation capability shortens processing time for drug registration documents and clinical trial documentation from days to minutes.

Enhance international competitiveness | Rapid generation of high-quality multilingual documents supports enterprise expansion into overseas markets.

Knowledge asset accumulation | Historical records form an enterprise-specific pharmaceutical translation corpus, which can feed back into terminology database optimization and future AI training.

3. Requirements

3.1 User Authentication and Data Isolation

- As a user, I want the system to prioritize Single Sign-On (e.g., Feishu) support so I can use unified corporate identity authentication without remembering multiple account credentials.

Acceptance Criteria:

1. The system login page should prominently display a "Feishu Login" button; clicking it redirects the user to the Feishu authorization page.

2. If Feishu SSO is unavailable due to configuration issues, the system must provide a backup username/password login form.

3. After successful login via either method, the user enters the main system interface with an active session that persists until manually logged out or timed out.

- As a user, I want the system to implement data isolation so that each user can only view and operate their own translation records, ensuring data security and privacy.

Acceptance Criteria:

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1. After User A logs in, the left-side history list displays only User A's translation tasks, with no records from other users shown.

2. User A cannot access or download User B's translation results by directly entering a URL or any other method.

3. The system must perform backend validation based on the logged-in user 's identity for all data operations to ensure data isolation and security.

3.2 General Language Configuration

- As a user, I want the system to support at least bidirectional translation between Chinese and English, and as many common languages as possible (e.g., Japanese, Korean, German, French), to handle multilingual translation scenarios.

Acceptance Criteria:

1. Above the input area in both text and file translation modes, there should be two dropdown menus for source language and target language, allowing users to freely select any supported language.

2. The source language dropdown must include options “ Auto-detect, ” “ Chinese, ” and “ English, ” with “ Auto-detect ” as the default. When the user inputs Chinese text, the system should automatically recognize it as “ Chinese ” ;

when English text is entered, it should automatically recognize it as “ English, ” and display the recognized result in the dropdown. The same applies to file uploads.

3. If the user manually changes the source language to a specific language (e.g., “ English ”), the system must use the user-selected language for translation and no longer override it automatically.

4. The target language dropdown does not support auto-detection; the user must manually select the target language. The enumeration must include at least “ Chinese ” and “ English. ”

3.3 Medical Industry General Terminology Database Mounting

- As a user, I want the system to call the medical industry general terminology database during translation to ensure accurate and consistent professional term translations.

Acceptance Criteria:

1. Upon submission of a translation request (text or file), the system backend must invoke the medical industry general terminology database to match and replace terms.

2. For medical terms included in the terminology database (e.g., "randomized controlled trial"), the translation must uniformly use the standard translation from the database (e.g., "randomized controlled trial")—no non-database literal translations are allowed.

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3. If a term is not found in the terminology database, the system shall use a general machine translation engine to process it normally, without interrupting the translation due to missing terms.

3.4 General Translation Requirements

As a user, I want to see a translation status prompt (e.g., "Loading", "Success", or "Failure") after clicking "Send", so I can understand the translation progress. When failure occurs, retry functionality should be supported.

Acceptance Criteria:

1. After the user clicks "Send", the button should immediately become grayed out, navigate to the result page, and display either "Translating..." or a loading animation to prevent repeated clicks.

2. Upon successful translation completion, the left side of the page should display the original text, the right side should show the translated text, and a "Translation Successful" message should appear briefly before automatically disappearing.

3. If translation fails due to network or server issues, an explicit "Translation Failed" message should be displayed, along with a "Retry" button in the translation area. When the user clicks "Retry", the system should re-initiate the translation request.

4. After a successful retry, the status should update to "Success"; if the retry still fails, the "Retry" button should be redisplayed again, with no limit on the number of retries.

3.3.1 Text Translation

As a user, I want to input medical text in text translation mode to quickly complete short texts or segments.

Acceptance Criteria:

1. When the user is in text translation mode, a multi-line text input box should appear on the right side, allowing the user to enter medical text up to a maximum of 5000 characters.

2. After the user inputs the text and clicks the "Send" button, the system should immediately begin processing the translation.

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3. During the translation process, the "Send" button should immediately be disabled and display "Translating..." or a loading animation until the translation is complete (whether successful or failed), at which point the button becomes usable again.

As a user, I want the translation result to appear directly on the right side of the page (not in a tooltip bubble) so that I can clearly view both the original text and the translated text, and copy the translation content.

Acceptance Criteria:

1. After text translation is completed, no tooltip-style bubble should appear on the right side of the page; instead, clear cards or areas should separately display "Original Text" and "Translated Text."

2. The original and translated texts must be clearly labeled (e.g., with headings "Original Text" and "Translated Text") for easy

comparison by the user.

3. Next to the translated text area, there should be a "Copy" button or icon.

4. When the user clicks the copy button, the translated text should be copied to the system clipboard, and a brief prompt saying "Copied" should appear.

5. The copied content must be complete and formatted as plain text without extra spaces or line breaks.

As a user, I want text translation to support only single-turn queries without maintaining multi-turn conversation context, so that I can focus on individual translation tasks without interference.

Acceptance Criteria:

1. After completing one text translation, if the user enters new text and clicks send again, the system should treat this as a completely new and independent task, with no contextual connection to previous original or translated text (e.g., it should not automatically refer to the subject of the previous sentence).

2. Each translation result should be displayed independently on the right side, without stacking or showing historical translations (historical records are saved separately in the left-side list).

3.3.2 File Translation

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3.4 Results Display and Download

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3.5 Historical Record Management

3.6 Interface and Experience

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3.3 Non-functional Objectives

Dimension | Objective

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Performance | Text translation response ≤ 2 seconds; File translation response ≤ 30 seconds (for documents up to 10 pages)

Availability | System annual availability $\geq 99.5\%$

Security | Data encrypted at rest and in transit using HTTPS; user sessions isolated

Compatibility | Supports latest two versions of Chrome and Edge; resolution 1280×720 or higher

Maintainability | Terminology database can be updated independently without releasing a new version